

Beau Dougan

San Diego, California

Portfolio: <https://beaudougan.com/>

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Education

California State University San Marcos

Graduated May 2026 Cum Laude

Bachelor of Science in Software Engineering

Relevant Coursework: Data Structures and Algorithms (**C++ & Java**), Operating Systems (**C++**), Software Testing (**Junit**), Embedded Systems (**Assembly & C**), Discrete Math, Software Engineering (**Java**), Database Systems (**SQL**), Software Architecture (**Java**), Computer Architecture, Machine Learning (**Python**)

Technical Skills:

- Languages: **Java, Assembly, C++, Python, SQL, Embedded C++**
- Tools/Technology: Git, VS Code, Unix, Junit, IntelliJ, Eclipse, SQL, AWS, Jenkins, scikit-learn, PostgreSQL

Relevant Work Experience:

Qualcomm Robotics Senior Design Project

September 2025 – May 2026

- Collaborated within a four-person Agile/SCRUM team to build an autonomous voice-activated guide robot using Python and ROS2, implementing LiDAR and IMU odometry using EKF for navigation and BLE for iOS device communication. This robot is able to move around a pregenerated 3D map created using SLAM Toolbox.
- Engineered requirements analysis, software integration, and testing, producing comprehensive technical documentation across the full engineering lifecycle. Shifting requirements and rapid prototyping required weekly meetings with a project manager at Qualcomm
- Made the decision to switch from visual SLAM using a Camera Pi 3 to LiDAR SLAM trading off affordability for accuracy

Projects:

Credit Card Fraud Detection

2026

- Created a machine learning model which can detect fraud with 96% recall using anonymized features from credit card purchases
- Used **scikit-learn** to build random forest and logistic regression models as well as comparing the two models

STM32 Door Access Simulator

2026

- Built a keypad-based access control system on an STM32 Nucleo board using Embedded C++, integrating PWM servo control, keypad scanning, digital I/O, and embedded state management to simulate door lock and unlock behavior with LED and buzzer feedback.

Additional Work Experience:

Doordash Delivery Driver

February 2024 - 2026

Mainstream Source Office Assistant

August 2022 - 2024

Sambazon Açaí Café Barista

August 2020 – 2022

Thai Society Delivery Driver

August 2017 – 2020

In-N-Out Burger

July 2014 – 2017